

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

This means traffic from R3 to the 192.168.2.0/24 network has to go through R1. The S0/0/1 interface on R2 is still set as passive, so R2 is not sending OSPF updates to R3 on that link. The total cost 129 comes from two serial links (64 + 64) plus the LAN interface cost (1)

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config) # router ospf 1

R2(config-router) # no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65.

It is 64 from ~~one~~ serial link plus 1 from the Gigabit LAN interface (64 + 1).

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

Because nowadays many links are faster than 100 Mbps. If we keep the default 100 Mbps reference bandwidth, all fast links (Fast Ethernet, Gigabit, etc.) will have the same cost. Changing the reference bandwidth makes the OSPF cost more accurate for modern high-speed networks.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

____ The OSPF cost is the sum of the outgoing interface costs along the path.

For 192.168.3.0/24, the cost is R1's serial interface cost plus R3's G0/0 cost ($781 + 1 = 782$).

For 192.168.23.0/30, the cost is the sum of the two serial interface costs along the path. Issue the bandwidth 128 command on all remaining serial interfaces in the topology. ($781 + 64 = 845$)

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

____ The new accumulated cost is 1562.

This is because each serial interface now has a cost of 781 (after setting bandwidth to 128 kbps), and the path includes two serial links ($781 + 781$).

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

Because after changing the interface bandwidth or cost, the path through R2 has a lower total OSPF cost compared to the direct path. OSPF always chooses the path with the lowest accumulated cost, so the route is updated to go through R2.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

The router ID uniquely identifies each router in the OSPF network. If it changes unexpectedly, neighbor relationships may reset and cause instability. Setting it manually makes the network more stable and predictable. _____

2. Why is the DR/BDR election process not a concern in this lab?

Because most links in this topology are point-to-point serial links. DR/BDR election only happens on multi-access networks like Ethernet. _____

3. Why would you want to set an OSPF interface to passive?

To stop sending OSPF hello packets on interfaces where no other routers exist, such as LANs connected to PCs. This reduces unnecessary OSPF traffic and improves security. _____

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

_____ OSPF uses areas. Dividing the network into areas makes it hierarchical and scalable. _____

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

_____ network 10.0.0.0 0.255.255.255 area 0 _____

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The EIGRP route will be used because it has the lowest administrative distance (90), compared to OSPF (110) and RIP (120). Administrative distance is considered before the routing metric. _____